THE INTELLIGENT HEALTHCARE COORDINATION NETWORK

# VITALIS

**Complete Technical Architecture & System Blueprint**

*Backend & Frontend Architecture • Algorithms • ML Model Integration • ITDC 2026*

## 1. Executive Overview & Core Philosophy

**VITALIS** addresses fragmentation across two dimensions of healthcare delivery: **External patient navigation** (matching patient needs to verified facility capabilities) and **Internal hospital coordination** (unifying Reception, Nursing, Doctor consultation, and Diagnostics).

> **Safety Invariant:** Patient Safety Before Automation. Emergency rules are deterministic. The AI/ML decision support layer provides non-diagnostic documentation and structured draft suggestions; it never autonomously diagnoses, prescribes, or overrules human clinical judgment.

## 2. Tiered System Architecture

The system is designed as a high-performance Modular Monolith:

| Tier | Components & Tech Stack | Key Responsibilities |
|---|---|---|
| **Presentation Tier** | Next.js 14+ App Router, TypeScript, Tailwind CSS, shadcn/ui | Patient PWA (triage, matching, QR consent, records); Hospital Portal (Reception, Nursing, Doctor, Diagnostics); Admin Console. |
| **API & Security Tier** | FastAPI, JWT Bearer Tokens, Argon2id, RBAC Dependencies | Role-scoped access control across 8 roles, request validation, rate limiting, and immutable audit event logging. |
| **Domain Services** | Encounter State Machine, ROUTE Matcher, CONNECT QR, TRANSFER Engine | Single persistent patient encounter; multi-criteria facility suitability; single-use tokenized QR grants; inter-hospital package exchange. |
| **Persistence Tier** | PostgreSQL / SQLite, SQLAlchemy ORM, Private Object Storage | Relational schema, versioned clinical notes, append-only audit trail, secure object storage for imaging & reports. |

## 3. Centralized Hospital Workflow: The Four Departments

A patient is registered once at Reception. Departmental staff contribute structured notes to the same encounter ID:

- **Reception (Intake):** Searches or registers patient; records complaint, arrival mode, and referral source; issues unique encounter number; routes to Nursing.
- **Nursing (Observations):** Records vital signs (BP, Pulse, Temp, SpO2, Resp Rate, Pain Score); automatic threshold alarms flag abnormal readings for urgent review.
- **Doctor (Consultation):** Consolidated view of intake and vitals; AI-assisted differential diagnosis drawer; orders diagnostic tests; signs consultation note.

- **Diagnostics (Lab & Radiology):** Receives orders in work queues; steps order through collection and result entry; authorized verification pushes results directly to doctor.

## 4. Mathematical Formulas & Algorithmic Engines

### 4.1 VITALIS ROUTE — Facility Matching Formula

The platform computes a transparent suitability score (0–100):
**S = 0.40·C + 0.25·A + 0.20·D + 0.15·V**

- **C (Service Compatibility - 40%):** Hard filter. Excludes any facility that cannot provide the essential service.
- **A (Reported Availability - 25%):** Available=100%, Limited=60%, Unknown=30%, Unavailable=0%. Unknown availability is penalized.
- **D (Distance Suitability - 20%):** Haversine great-circle decay (<=2km: 100%, 10km: 62%, 30km: 22%).
- **V (Verification & Freshness - 15%):** Verified=100%, Pending=60%. Decays if confirmed >24h (-15%) or >48h (-30%).

### 4.2 VITALIS CONNECT — Cryptographic QR Identity

The QR code encodes a high-entropy random single-use token URL. The server stores only the **SHA-256 hash** of the token. **Zero medical data or patient PII is stored inside the QR code.** Grants expire within 5–10 minutes and support instant revocation.

## 5. Frontend Architecture (Next.js 14+ PWA)

Built as a mobile-first Progressive Web App (PWA) using Next.js 14 App Router, Tailwind CSS, and shadcn/ui. It supports offline static caching and optimistic client-side drafts for low-connectivity environments.

| Route Group | User Target | Key Capabilities |
|---|---|---|
| **/patient/*** | Patient PWA | Symptom intake wizard, nearby facility search, consent and QR sharing, longitudinal records, referral tracker. |
| **/hospital/*** | Clinical Staff | Reception intake desk (/reception), Nursing vital triage (/nursing), Doctor consultation console (/doctor), Diagnostics queue (/diagnostics). |
| **/admin/*** | Hospital Admin | Facility service availability management, staff role permissions, immutable audit log inspection. |

## 6. Machine Learning Model Integration Guide

The backend provides a standardized VitalisMLModelAdapter slot in app/services/intelligence.py. When your ML model is ready, simply hook its predictions into this adapter. Deterministic safety red flags run first; then the model generates differential candidates, requiring mandatory physician review.

## 7. Verified Demo Accounts (Password: vitalis2026)

All 7 automated test suites are passing. The demo database is seeded with realistic Abeokuta healthcare data:

| Department / Role | Login Email | Affiliation / Context |
|---|---|---|
| Reception | reception@fmc.vitalis.ng | FMC Abeokuta (Front Desk) |
| Nursing | nurse@fmc.vitalis.ng | FMC Abeokuta (Triage Station) |
| Doctor | doctor@fmc.vitalis.ng | FMC Abeokuta (Clinics) |
| Lab Tech | lab@fmc.vitalis.ng | FMC Abeokuta (Clinical Pathology) |
| Radiologist | rad@fmc.vitalis.ng | FMC Abeokuta (Imaging Dept) |
| Hospital B Doctor | doctor@statehospital.vitalis.ng | State Hospital Sokenu (Referral recipient) |
| Patient | olumide@example.com | Olumide Adeleke (VIT-2026-0001) |